INCH-POUND
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COMMERCIAL ITEM DESCRIPTION

SHIRT, LIQUID COOLING, LONG SLEEVE

The General Services Administration has authorized the use of this Commercial Item Description for all Federal agencies.

1. SCOPE. This commercial item description covers the requirements for a liquid cooling shirt intended to be worn by aircrew and other defense personnel as part of a microclimate cooling system. The shirt cools the user when a liquid coolant at a temperature lower than skin temperature is pumped through it. The garment is worn as a shirt under outer protective clothing, directly against the skin.

2. CLASSIFICATION. The cooling shirt shall be of one type in the following sizes:

Small
Medium
Large

3. SALIENT CHARACTERISTICS

3.1 Description. The cooling shirt shall have long sleeves with cuffs, crew neck collar, waistband, and slide fastener front (right hand pull) with a protective flap behind the slide fastener. All seams shall be safety stitched. The cooling garment shall contain a network of tubing attached to the inside of the shirt. When the shirt is worn, the tubing shall be in contact with the skin to enable maximum heat transfer to occur. The cooling tubing shall be arranged so that the tubing does not kink when arms are bent or when bending over or twisting at the torso. Inlet and outlet tubing shall exit to the right from the bottom front center of the cooling shirt. The main network tubing shall be attached onto the inside surface of the cooling shirt. The cooling shirt shall be capable of directly cooling the front and back of the torso and both arms. The weight of the cooling shirt shall not exceed 2.25 pounds dry (empty) and 3.5 pounds wet (full).

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Defense Personnel Support Center, Clothing and Textiles Directorate, Attn: DPSC-FNS, 2800 South 20th Street, Philadelphia, PA 19145-5099.

AMSC N/A **FSC 8415**

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The joints between the tubing connectors (manifolds, tees, elbows, etc.) and flexible tubing shall be chemically or ultrasonically welded. A repair kit shall be available from the shirt manufacturer to permit permanent repair of the shirt.

3.2 Materials

3.2.1 Basic material. The basic material shall be an interlock knit, natural color, 100 percent meta-aramid fabric (1.1 Dtex or 1 Denier) or a fiber with equivalent properties and 6 ± 0.5 ounces per square yard when tested in accordance with ASTM D3776 Opt C. The material shall meet or exceed the flame resistance requirements of not more than 1.0 second average time of after flame and a char length of not more than 2.5 inches in both the warp and filling directions when tested in accordance with method 5903 of FED-STD-191. The fabric shall have a minimum of 34 courses per inch and a minimum of 28 wales per inch when tested in accordance with ASTM D3887.

3.2.2 Thread. The natural color thread shall be texturized continuous multifilament polyester, commercial size 35-45, at 10,000 yards per pound, with 2.9 pound breaking strength (minimum) at 35% elongation (maximum) when tested in accordance with ASTM D2256. A commercial thread that meets the above requirements is also suitable. Or as an alternative the thread shall be intermediate modulus, normal spun, para-aramid, which shall not carbonize at a temperature below 6750 F. The thread shall be 1.5 denier per filament, 2 ply, 11,970 to 13,230 yards per pound, with a breaking strength of 6.0 pounds and maximum elongation of 6% when tested in accordance to ASTM D2256. A commercial thread that meets the above requirements is also suitable.

The overcast (overedge) threads which fasten the tubing network to the cooling shirt shall be continuous filament polyester, 500 yards per pound, with a minimum breaking strength of 3.2 pounds, and a maximum elongation of 35% when tested in accordance with ASTM D2256. The thread shall contain only the minimum amount of lubricant to facilitate sewing. The yarn finish shall be nonstaining and non flame propagating.

3.2.3 Main cooling tubing network. The tubing network shall be comprised of clear Poly Vinyl Chloride (PVC) tubing. The network tubing size shall be 3/32 inch nominal I.D. (inside diameter) x 5/32 inch nominal O.D. (outside diameter). The network tubing shall be stitched to the inside of the shirt at 12-14 stitches per inch, all other seams shall be overedged 14 to 18 stitches per inch. The distribution of cooling tubing in the shirt shall be as follows: 30 ± 5% in the arms (distributed equally among both arms), 30 ± 5% on the front of the shirt (distributed equally on both sides of the slide fastener), and 40 ± 5 % on the back of the shirt. The linear quantity of tubing per square foot shall be measured with the shirt lying flat in the relaxed state. The tubing network shall be as evenly distributed as practical. The network shall consist of a minimum of seven separate continuous circuits or loops of approximately equal length beginning and ending at the supply and return lead tubing, respectively. Each arm shall be served by a minimum of two individual circuits. As a minimum, the "small" shirt shall contain 140 feet of cooling tubing, the "medium" shirt shall contain 160 feet of cooling tubing, and the "large" shirt

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shall contain 180 feet of cooling tubing. Paragraph 3.6.2 describes cooling tubing length verification methods.

3.2.4 Lead tubing. The manifold lead tubing is the inlet and outlet tubes for the garment. The manifold lead tubing shall be a pair of 3/16 inch I.D. x 5/16 inch O.D. clear PVC tubing. The lead tubing shall run a minimum of 24 inches free of the shirt. This lead tubing shall exit the garment from the bottom front center of the shirt within 3 inches of the bottom of the shirt. The lead tubing shall exit to the right from the left side (as worn) of the shirt (within 3 inches of the slide fastener).

3.2.5 Permanent Repair Kit. A permanent repair kit shall be available from the cooling shirt manufacturer. The repair kit shall contain all materials and tools required to mend damaged tubing and fabric.

3.2.6 Slide fastener. The slide fastener shall be commercial size 5-7, separating, continuous element with a locking slider. It shall have a chain with crosswise breaking strength of 130 pounds (minimum) when tested in accordance with ASTM D2061. The color shall be black, white, or natural.

3.2.7 Slide fastener flame flap. A flame resistant flap running the full length of the slide fastener shall be located directly behind the slide fastener. The flame resistant flap material shall meet all criteria as described in 3.2.1. The flame resistant flap shall not interfere with the operation of the slide fastener.

3.3 Labels. Each cooling shirt shall have a size label and a commercial type identification/ instruction label. The identification/instruction label shall be located on the inside lower side seam and shall contain the item name, contractor's name, contract number, national stock number (NSN), fiber content and laundering information. The color of the labels shall be white.

3.3.1 Size label. The label shall include the size designation. The size label shall be located inside the garment at the center back of the collar and neck joining seam. The lettering for the size shall be 1/4 + 1/16 inch. The letter "L" shall be used for a large garment, the letter "M" shall be used for a medium size garment, and the letter "S" shall be used for a small size garment.

3.3.2 Label/tag. Each item shall be individually bar-coded with a tag for personal clothing items. The nonpermanent tags shall be attached to each item, and shall be clearly legible and readable by a scanner. The bar coding element shall be a 13 digit national stock number (NSN). The bar code type shall be a medium to high code density and shall be located so that it is completely visible on the item when it is folded and/or packaged as specified.

3.4 Finished Measurements. The cooling shirt shall fit snugly against the skin but shall not hinder mobility or movement. When a person wearing the proper size garment is seated, the tubing will not fold and/or kink at the waist. The cooling shirt shall be measured laying flat and smooth, with the front slide fastener completely closed. The cooling shirt shall conform to the measurements in Table I.

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TABLE I. FINISHED MEASUREMENTS (INCHES)

Garment Dimension	S	M	L	Tolerance
One half chest 1/	15	17	19	±1/2
Back length 2/	22-1/2	24	25-1/2	+1, -0
Front length 3/	23-1/2	25	26-1/2	±1/2
Sleeve length 4/	18	19	20	+3/4, -0
One half neck 5/	8	8-1/2	9	±1/2
Slide fastener length	20	21	22	±1
One half cuff width 6/	3-1/4	3-1/4	3-1/4	± 1/4
Cuff height	1	1	1	± 1/4
Crew neck collar height	1	1	1	± 1/4
Waistband height	1-1/2	1-1/2	1-1/2	± 1/4

1/ Measure at base of armhole, from folded edge to folded edge.
2/ Measure along center back from collar seam to bottom of shirt.
3/ Measure from top of shoulder seam at neck to the bottom of the garment front.
4/ Measure along underarm seam from bottom of cuff to armhole seam.
5/ Measure from collarette seam to opposite collarette seam along the curved edge.
6/ Measure at center of cuff from folded edge to folded edge.

3.5 Performance. The pressure drop across the cooling shirt's network tubing shall not exceed 10 (psig) at 10 gallons per hour (gph) flow of coolant. The tubing network shall be designed such that heat removal during operation is evenly distributed over all major areas of the upper body in contact with the cooling garment.

3.6 Testing and verification

3.6.1 Leak testing. Connect the cooling garment to a compressed air source or water source regulated at 40 pounds per square inch (psig). Pressurize the cooling shirt tubing to 40 psig for 3 minutes. Inspect all cooling network tubing and manifolds closely for leaks and/or deformation. Any leaks and/or deformation are grounds for rejection of the cooling shirt. Reduce pressure to 20 psig for 2 minutes. If pressure in the shirt drops below 18 psig at the end of two minutes, it shall be rejected.

3.6.2 Cooling tubing length verification. The cooling tubing network shall be measured for total length exclusive of the lead tubing and the manifold system. Suggested methods of measurement are as follows: 1) measure directly using a ruler or other suitable measuring device; 2) determine volume of water that the shirt's tubing can contain then convert to a linear quantity of tubing using the known dimensions of tubing (Note: Ensure that the effects of the lead tubing and manifold system are not included in the measurement); 3) weigh the shirt in both the full (wet) condition and the empty (dry) condition. Translate the difference in weight to the linear quantity of tubing. (Note: ensure that the effects of the lead tubing and manifold system are not included in the measurement.) The measured length shall comply with the minimum lengths cited in 3.2.3.

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3.6.3 Flow testing/pressure drop. The cooling shirt shall be tested for flow/pressure drop by connecting the shirt's inlet tube to a pump capable of pumping water at a rate of 0 to 12 gallons per hour (gph) at 10 pounds per square inch gage (psig). A pressure transducer or gage capable of measuring gage pressure is to be installed on the inlet lead tube between the pump and the garment. The outlet (return) lead tube is to expel water at atmospheric pressure. The pressure drop through the garment at a flow rate of 10 gallons per hour shall be no more than 10 psig (Note: Flow testing/pressure drop testing shall not include the lead tubing end connectors).

3.6.4 Shirt size evaluation. Shirts of each of the three sizes shall be measured against the finished measurements as stated in Table I.

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specification, standards and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.1.2 Defects. Any component not in accordance with specified requirements, any hole, cut, or tear; color deviations not as specified; any part shaded; spotted or stained (outside); raw edges, open seams, thread ends, or loose tension resulting in loose seams; leaking and deformed tubing and manifolds; tubing length not as specified; any serious material defects, distorted parts or other evidence of workmanship; any component part omitted; label missing, incorrect, or illegible; measurement of item not as specified; bar code omitted or not readable by scanner; human-readable interpretation (HRI) omitted or illegible; bar code not visible on folded, packaged item; bar code causes damage to the item; any items not packaged in accordance with the contract or purchase order.

5.2 Acceptance criteria. Acceptance criteria shall be as specified in the contract or purchase order.

6. PACKAGING

6.1 Preservation, packing, and marking. The preservation, packing, and marking shall be as specified in the contract or order.

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7. NOTES

7.1 Source of Government documents. Copies of military and Federal documents are available from:

Standardization Documents Order Desk
Bldg. 4D
700 Robbins Avenue
Philadelphia, PA 19111-5094

7.2 Source of non-Government documents

AATCC Documents

(Applications for copies should be addressed to American Association of Textile Chemists and Colorists (AATCC), P.O. Box 12215, Triangle Park, NC 27709-2215.)

MILITARY INTERESTS:

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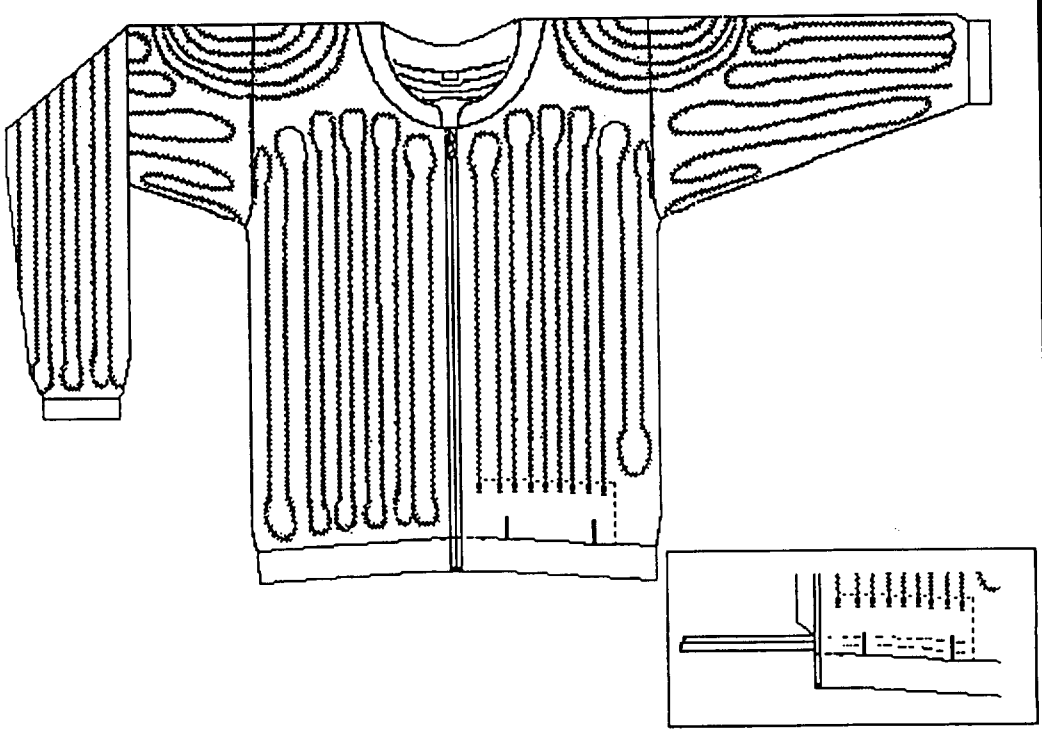
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PREPARING ACTIVITY:
DLA - CT

Project 8415-0071

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NOTE: Figure should not be used for tubing placement

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